



Prayas JEE 2024

Motion In A Plane (PYQ)

1. An object is thrown vertically upwards. At its maximum height, which of the following quantity becomes zero? **[JEE Main 2022]**
(A) Momentum (B) Potential energy
(C) Acceleration (D) Force
2. A ball is projected vertically upward with an initial velocity of 50 ms^{-1} at $t = 0\text{s}$. At $t = 2\text{s}$, another ball is projected vertically upward with same velocity. At $t = \underline{\hspace{1cm}}$ s, second ball will meet the first ball ($g = 10 \text{ ms}^{-2}$) **[JEE Main 2022]**
3. A body is projected from the ground at an angle of 45° with the horizontal. Its velocity after 2s is 20 ms^{-1} . The maximum height reached by the body during its motion is $\underline{\hspace{1cm}}$ m. (use $g = 10 \text{ ms}^{-2}$) **[JEE Main 2022]**
4. A fighter jet is flying horizontally at a certain altitude with a speed of 200 ms^{-1} . When it passes directly overhead an anti-aircraft gun, bullet is fired from the gun, at an angle θ with the horizontal, to hit the jet, if the bullet speed is 400 m/s , the value of θ will be $\dots\dots\dots^\circ$ **[JEE Main 2022]**
5. A projectile is launched at an angle ' α ' with the horizontal with a velocity 20 ms^{-1} . After 10 s , its inclination with horizontal is ' β '. The value of $\tan\beta$ will be: ($g = 10 \text{ ms}^{-2}$) **[JEE Main 2022]**
(A) $\tan\alpha + 5 \sec \alpha$ (B) $\tan\alpha - 5 \sec \alpha$
(C) $2 \tan\alpha - 5 \sec \alpha$ (D) $2 \tan\alpha + 5 \sec \alpha$
6. A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is : **[JEE Main 2022]**
(A) 30 kmh^{-1} (B) $\frac{25}{\sqrt{2}} \text{ kmh}^{-1}$
(C) $\frac{30}{\sqrt{2}} \text{ kmh}^{-1}$ (D) 25 kmh^{-1}
7. Motion of particle in x-y plane is described by a set of following equations $x = 4 \sin \left(\frac{\pi}{2} - \omega t \right) \text{m}$ and $y = 4 \sin (\omega t) \text{m}$. The path of particle will be **[JEE Main 2022]**
(A) Circular (B) Helical
(C) parabolic (D) elliptical
8. A person can throw a ball upto a maximum range of 100 m . How high above the ground he can throw the same ball? **[JEE Main 2022]**
(A) 25 m (B) 50 m
(C) 100 m (D) 200 m
9. A particle is moving in a straight line such that its velocity is increasing at 5 ms^{-1} per meter. The acceleration of the particle is $\underline{\hspace{1cm}} \text{ ms}^{-2}$ at a point where its velocity is 20 ms^{-1} . **[JEE Main 2022]**
10. Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their range respectively will be **[JEE Main 2022]**
(A) $1 : \sqrt{2}$ (B) $\sqrt{2} : 1$
(C) $2 : \sqrt{3}$ (D) $\sqrt{3} : 2$
11. A ball is thrown vertically upwards with a velocity of 19.6 ms^{-1} from the top of a tower. The ball strikes the ground after 6 s . The height from the ground up to which the ball can rise will be $\left(\frac{k}{5} \right) \text{m}$. The value of k is $\underline{\hspace{1cm}}$ (use $g = 9.8 \text{ m/s}^{-2}$) **[JEE Main 2022]**



12. A ball is projected from the ground with a speed 15ms^{-1} at an angle θ with horizontal so that its range and maximum height are equal, then ' $\tan \theta$ ' will be equal to **[JEE Main 2022]**
(A) $1/4$ (B) $2 : 1$
(C) 2 (D) 4
13. Two projective thrown at 30° and 45° with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is **[JEE Main 2022]**
(A) $1 : \sqrt{2}$ (B) $2 : 1$
(C) $\sqrt{2} : 1$ (D) $1 : 2$
14. If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y = 5x(1 - x)$. The y component vector of the initial velocity is _____ $\hat{j}$. **[JEE Main 2022]**
(Take $g = 10\text{m/s}^2$)
15. A body of mass 10 kg is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point $(20, 10)$. If T is the time of flight, then the momentum vector, at time $t = \frac{T}{\sqrt{2}}$, is _____ (Take $g = 10\text{ m/s}^2$) **[JEE Main 2022]**
(A) $100\hat{i} + (100\sqrt{2} - 200)\hat{j}$
(B) $100\sqrt{2}\hat{i} + (100 - 200\sqrt{2})\hat{j}$
(C) $100\hat{i} + (100 - 200\sqrt{2})\hat{j}$
(D) $100\sqrt{2}\hat{i} + (100\sqrt{2} - 200)\hat{j}$
16. A ball is projected with kinetic energy E , at an angle of 60° to the horizontal. The kinetic energy of this ball at the highest point of its flight will become : **[JEE Main 2022]**
(A) Zero (B) $E/2$
(C) $E/4$ (D) E
17. An object is projected in the air with initial velocity u at an angle θ . The projectile motion is such that the horizontal range R , is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity of the angle of projection, at which the second object is projected, will be _____ degree. (Mark the smallest angle possible) **[JEE Main 2022]**
18. A person is swimming with a speed of 10 m/s at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is ' x ' m/s . The value of ' x ' to the nearest integer is _____. **[JEE Main 2021]**
19. A force $\vec{F} = (40\hat{i} + 6\hat{j})\text{N}$ acts on a body of mass 5 kg . If the body starts from rest, its position vector $\vec{r}$ at time $t = 10\text{ s}$, will be : **[JEE Main 2021]**
(A) $(100\hat{i} + 400\hat{j})\text{m}$ (B) $(100\hat{i} + 100\hat{j})\text{m}$
(C) $(400\hat{i} + 100\hat{j})\text{m}$ (D) $(400\hat{i} + 400\hat{j})\text{m}$
20. A swimmer wants to cross a river from point A to point B. Line AB makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer is same as that of the river. The angle θ with the line AB should be _____°, so that the swimmer reaches point B. **[JEE Main 2021]**
21. A bomb is dropped by fighter plane flying horizontally. To an observer sitting in the plane, the trajectory of the bomb is a : **[JEE Main 2021]**
(A) hyperbola
(B) parabola in the direction of motion of plane
(C) straight line vertically down the plane
(D) parabola in a direction opposite to the motion of plane
22. If the velocity of a body related to displacement x is given by $v = \sqrt{5000 + 24x}\text{ m/s}$, then the acceleration of the body is m/s^2 . **[JEE Main 2021]**



23. A player kicks a football with an initial speed of 25 ms^{-1} at an angle of 45° from the ground. What are the maximum height and the time taken by the football to reach at the highest point during motion? (Take $g = 10 \text{ ms}^{-2}$) [JEE Main 2021]
- (A) $h_{\max} = 10 \text{ m}$ $T = 2.5 \text{ s}$
(B) $h_{\max} = 15.625 \text{ m}$ $T = 3.54 \text{ s}$
(C) $h_{\max} = 15.625 \text{ m}$ $T = 1.77 \text{ s}$
(D) $h_{\max} = 3.54 \text{ m}$ $T = 0.125 \text{ s}$

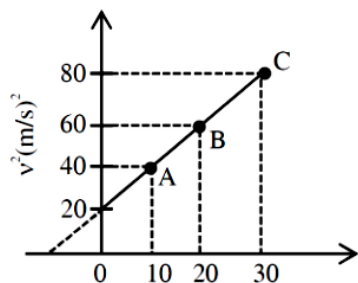
24. A helicopter is flying horizontally with a speed ' v ' at an altitude ' h ' has to drop a food packet for a man on the ground. What is the distance of helicopter from the man when the food packet is dropped?

[JEE Main 2021]

- (A) $\sqrt{\frac{2ghv^2 + 1}{h^2}}$ (B) $\sqrt{2ghv^2 + h^2}$
(C) $\sqrt{\frac{2v^2h}{g} + h^2}$ (D) $\sqrt{\frac{2gh}{v^2} + h^2}$

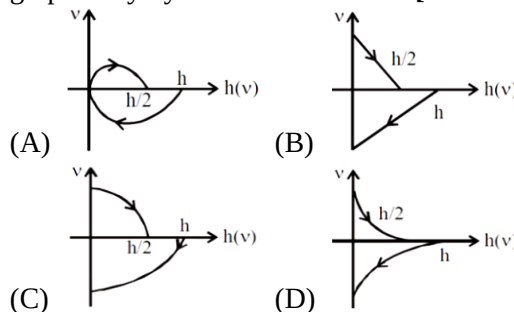
25. A particle is moving with constant acceleration ' a '. Following graph shows v^2 versus x (displacement) plot. The acceleration of the particle is $\underline{\hspace{1cm}} \text{ m/s}^2$.

[JEE Main 2021]



26. Starting from the origin at time $t = 0$, with initial velocity $5\hat{j} \text{ ms}^{-1}$, a particle moves in the x - y plane with a constant acceleration of $(10\hat{i} + 4\hat{j}) \text{ ms}^{-2}$. At time t , its coordinates are $(20 \text{ m}, y_0 \text{ m})$. The values of t and y_0 , are respectively: [JEE Main 2020]
- (A) 4 s and 52 m (B) 2 s and 24 m
(C) 2 s and 18 m (D) 5 s and 25 m

27. A Tennis ball is released from a height h and after freely falling on a wooden floor it rebounds and reaches height $\frac{h}{2}$. The velocity versus height of the ball during its motion may be represented graphically by : [JEE Main 2020]



28. When a car is at rest, its driver sees rain drops falling on it vertically. When driving the car with speed v , he sees that rain drops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to $(1 + \beta)v$, this angle changes to 45° . The value of β is close to: [JEE Main 2020]

- (A) 0.41 (B) 0.50
(C) 0.37 (D) 0.73

29. A particle starts from the origin at $t = 0$ with an initial velocity of $3.0\hat{i} \text{ m/s}$ and moves in the x - y plane with a constant acceleration $(6.0\hat{i} + 4.0\hat{j}) \text{ m/s}^2$. The x -coordinate of the particle at the instant when its y -coordinate is 32 m is D meters. The value of D is:

[Main 9 Jan. 2020 (II)]

- (A) 32 (B) 50
(C) 60 (D) 40

30. A particle moves such that its position vector $\vec{r}(t) = \cos \omega t \hat{i} + \sin \omega t \hat{j}$ where ω is a constant and t is time. Then which of the following statements is true for the velocity $\vec{v}(t)$ and acceleration $\vec{a}(t)$ of the particle: [Main 8 Jan. 2020 (II)]

- (a) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed away from the origin
(b) $\vec{v}$ and $\vec{a}$ both are perpendicular to $\vec{r}$
(c) $\vec{v}$ and $\vec{a}$ both are parallel to $\vec{r}$
(d) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed towards the origin



31. The position vector of a particle changes with time according to the relation $\vec{r}(t) = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$.

What is the magnitude of the acceleration at $t = 1$?

- (A) 40 (B) 25
(C) 100 (D) 50

32. A particle moves from the point $(2.0\hat{i} + 4.0\hat{j})\text{m}$, at $t = 0$, with an initial velocity $(5.0\hat{i} + 4.0\hat{j})\text{ms}^{-1}$. It is acted upon by a constant force which produces a constant acceleration $(4.0\hat{i} + 4.0\hat{j})\text{ms}^{-2}$. What is the distance of the particle from the origin at time 2s?

[Main 11 Jan. 2019 (II)]

- (a) 15 m (b) $20\sqrt{2}$ m
(c) 5 m (d) $10\sqrt{2}$ m

33. A particle is moving along the x-axis with its coordinate with time 't' given by $x(t) = 10 + 8t - 3t^2$. Another particle is moving along the y-axis with its coordinate as function of time given by $y(t) = 5 - 8t^3$. At $t = 1$ s, the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. then v (in m/s) is _____. [Main 8 Jan. 2020 (I)]

34. The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle θ_0 with speed v_0 then ($g = 10 \text{ ms}^{-2}$)

[Main 12 Apr. 2019 (I)]

- (A) $\theta_0 = \sin^{-1} \frac{1}{\sqrt{5}}$ and $v_0 = \frac{5}{3} \text{ ms}^{-1}$
(B) $\theta_0 = \cos^{-1} \left(\frac{2}{\sqrt{5}} \right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$
(C) $\theta_0 = \cos^{-1} \left(\frac{1}{\sqrt{5}} \right)$ and $v_0 = \frac{9}{3} \text{ ms}^{-1}$
(D) $\theta_0 = \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$

35. Two gun A and B can fire bullets at speed 1 km/s and 2 km/s respectively. From a point on a horizontal ground, they are fired in all possible directions. The ratio of maximum areas covered by the bullets fired by the two guns, on the ground is:

[Main 10 Jan. 2019 (I)]

- (A) 1 : 16 (B) 1 : 2
(C) 1 : 4 (D) 1 : 8

36. The position of a projectile launched from the origin at $t = 0$ is given by $\vec{r} = (40\hat{i} + 50\hat{j})\text{m}$ at $t = 2\text{s}$. If the projectile was launched at an angle θ from the horizontal, then θ is (take $g = 10 \text{ ms}^{-2}$)

[Main Online April 9, 2014]

- (a) $\tan^{-1} \frac{2}{3}$ (b) $\tan^{-1} \frac{3}{2}$
(c) $\tan^{-1} \frac{7}{4}$ (d) $\tan^{-1} \frac{4}{5}$

37. A projectile is given a initial velocity of $(\hat{i} + 2\hat{j})\text{m/s}$, where $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. If $g = 10 \text{ m/s}^2$, the equation of its trajectory is

[Main 2013]

- (A) $y = x - 5x^2$ (B) $y = 2x - 5x^2$
(C) $4y = 2x - 5x^2$ (D) $4y = 2x - 25x^2$

38. A ball projected from ground at an angle of 45° just clears a wall in front. If point of projection is 4 m from the foot of wall and ball strikes the ground at a distance of 6 m on the other side of the wall, the height of the wall is :

[Main Online April 22, 2013]

- (A) 4.4 m (B) 2.4 m
(C) 3.6 m (D) 1.6 m

39. The stream of a river is flowing with a speed of 2 km/h. A swimmer can swim at a speed of 4 km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight?

[Main 9 April 2019 (I)]

- (A) 90° (B) 150°
(C) 120° (D) 60°

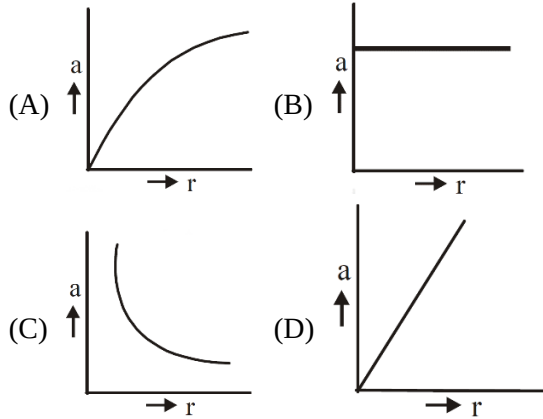
40. A particle is moving along a circular path with a constant speed of 10 ms^{-1} . What is the magnitude of the change in velocity of the particle, when it moves through an angle of 60° around the center of circle?

[Main Online April 10, 2015]

- (A) $10\sqrt{3} \text{ m/s}$ (B) zero
(C) $10\sqrt{2} \text{ m/s}$ (D) 10 m/s



41. If a body moving in circular path maintains constant speed of 10 ms^{-1} , then which of the following correctly describes relation between acceleration and radius? **[Main Online April 10, 2015]**



42. A boat which has a speed of 5 km/hr in still water crosses a river of width 1 km along the shortest possible path in 15 minutes. The velocity of the river water in km/hr is **[1998 – 1 Mark]**

(A) 1 (B) 3
(C) 4 (D) $\sqrt{41}$

43. A river is flowing from west to east at a speed of 5 meters per minute. A man on the south bank of the river, capable of swimming at 10 metres per minute in still water, wants to swim across the river in the shortest time. He should swim in a direction **[1983 – 1 Mark]**

(A) due north (B) 30° east of north
(C) 30° west of north (D) 60° east of north



Answer Key

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|-----------|-----------|
| 1. (A) | 23. (C) |
| 2. (6) | 24. (C) |
| 3. (20) | 25. (1) |
| 4. (60) | 26. (C) |
| 5. (B) | 27. (C) |
| 6. (C) | 28. (D) |
| 7. (A) | 29. (C) |
| 8. (B) | 30. (D) |
| 9. (100) | 31. (D) |
| 10. (C) | 32. (B) |
| 11. (392) | 33. (580) |
| 12. (D) | 34. (C) |
| 13. (C) | 35. (A) |
| 14. (5) | 36. (C) |
| 15. (D) | 37. (B) |
| 16. (C) | 38. (B) |
| 17. (15) | 39. (C) |
| 18. (5) | 40. (D) |
| 19. (C) | 41. (C) |
| 20. (30) | 42. (B) |
| 21. (C) | 43. (A) |
| 22. (12) | |